



Published in final edited form as:

Cogn Syst Res. 2019 June ; 55: 192–204. doi:10.1016/j.cogsys.2019.02.003.

Effects of Task Constraint on Action Dynamics

Patric C. Nordbeck^{1,*}, Laura K. Soter², Johan S. Viklund³, Emily A. Beckmann¹, Rachel W. Kallen⁴, Anthony P. Chemero^{1,5}, and Michael J. Richardson⁴

¹Department of Psychology, Center for Cognition, Action, & Perception, Perceptual Motor-Dynamics Lab, University of Cincinnati, Cincinnati, OH, USA

²Department of Philosophy, University of Michigan, Ann Arbor, MI, USA

³Faculty of Health Sciences, Cultural Selection and Behavioral Economics Lab, Oslo Metropolitan University, Oslo, Norway

⁴Department of Psychology and Center for Elite Performance, Expertise and Training, Macquarie University, Sydney, NSW, Australia

⁵Department of Philosophy, University of Cincinnati, Cincinnati, OH, USA

Abstract

The actualization of action possibilities (i.e., affordances) can often be accomplished in numerous ways. For instance, an individual could walk over to a rubbish bin to drop an item in or throw the piece of rubbish into the bin from some distance away. The aim of the current study was to investigate the action dynamics that emerge from such under-constrained task or action spaces using an object transportation task. Participants were instructed to transport balls between a starting location and a large wooden box located 9 meters away. The temporal interval between the sequential presentation of balls was manipulated as a control parameter and was expected to influence the distance participants moved prior to throwing or dropping the ball into the target box. A two-parameter state space derived from the Cusp Catastrophe Model was employed to illustrate how behavioral variability emerged as a consequence of the under-constrained task context. Two follow-up experiments demonstrated direct correspondence between model predictions and observed action dynamics as a function of increasing task constraints. Implications for modelling, the theory of affordances, and empirical studies more generally are discussed.

Keywords

affordances; dynamic systems; cusp catastrophe; dynamic modeling; under-constrained

* Author note: All correspondence should be addressed to Patric Nordbeck (nordbecp@mail.uc.edu), UC A&S Psychology Department, 47 W. Corry Blvd, Edwards 1 Bldg Suite 4150, Cincinnati OH 45219-001, United States of America.

Publisher's Disclaimer: This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final citable form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Declarations of interest: none.

Some of the data was published in the Proceedings of the 39th Cognitive Science Society and presented at Cognitive Science Society and Society for Chaos Theory in Psychology and the Life Sciences conferences, both in 2017.

1. Effects of Task Constraint on Action Dynamics

Reorganizing one's behavior in relation to changing task demands is a ubiquitous aspect of everyday life and is often required to ensure task success. In order to solve everyday tasks, human (and animal) behavior is functionally organized and reorganized in relation to the action possibilities, or affordances, that define a given environmental task context. The term *affordance* refers to action possibilities that emerge within an organism-environment system (Gibson, 1979; Chemero, 2009), and highlights the mutual reciprocity between relevant aspects of an organism and its environment. That is, affordances point both to the organism and the environment, relationally defining the fit between the two. For example, an individual concerned with a place to sit needs an environmental surface that is not only able to support their weight, but is also of an appropriate height relative to their leg length. There are of course many surfaces that are located too low or too high, or that cannot support a human's weight, in which case that particular surface does not afford sitting.

As with detecting sit-able surfaces, animal-environment relations are central to characterizing the climb-ability of stair risers: one's ability to climb a riser depends on the riser's height compared to one's leg length. Warren (1984) investigated this relationship by having participants judge if they perceived themselves capable of climbing a riser or not. The results revealed that participants judged the climb-ability of risers based on intrinsic, body-scaled information (i.e., riser height with respect to leg length) and perceived the boundary between climbable and unclimbable risers at an invariant riser height to leg length ratio of .85. Similar body-scaled ratios are known to define the perceived boundaries of a wide range of action possibilities, from the stand-ability and sit-ability of surfaces (e.g. Fitzpatrick, Carello, Schmidt & Corey, 1994; Mark, 1987), to the pass-through-ability of apertures (Warren & Whang, 1987), and the reach-ability and grasp-ability of objects (e.g. Carello, Groszofsky, Reichel & Solomon, 1989; Cesari & Newell, 1999; Richardson, Marsh, Baron, 2007). In each case, individuals correctly detect *affordance boundaries* (i.e. the boundary between when an action is or is not possible) by means of intrinsic body-scaled information and organize or reorganize their behavioral activity accordingly (Carello et al., 1989).

Of special relevance here is that affordance research has also illustrated how individuals typically exhibit *non-linear transitions* between different modes of affordance actualization. For instance, Richardson et al. (2007) explored the transition between one- and two-handed grasping by scaling the length of planks relative to hand length. The results revealed that participants (on average) exhibited a non-linear transition from one-handed grasping (for shorter planks) to two-handed grasping (for longer planks) at a plank-length to hand-span ratio of approximately .75. Figure 1 (*Left*) illustrates the kind of non-linear behavioral mode transitions observed in the study (also see van der Kamp & Savelsbergh, 1998), whereby an individual typically transitions from one mode of behavior (e.g., one-hand grasping) to a second mode of behavior (i.e., two-hand grasping) at a critical body- or action-scaled ratio. Similar non-linear phase transitions have been observed with respects to the affordances of striking in boxing (Hristovski, Davids, & Araújo, 2006), reaching (Carello et al., 1989), and grasping (Lopresti-Goodman, Turvey, & Frank, 2013).

For the present study, Figure 1 (*left*) depicts a generalized state or behavioral mode space, where two orthogonal modes of behavior are defined in terms of the probability (or proportion of time) that a particular state or behavioral mode (in this case Action Mode 1) would be expected to be observed or realized with respect to the value of a task relevant control parameter. The middle, grey area is of particular interest, as it defines where a behavioral mode transition between Action Mode 1 and some other action mode (i.e., Action Mode 2) or the elimination of Action Mode 1 is expected to occur. Here, we refer to this space as the ‘*transition space*’ and in a typical affordance experiment like those described above, this transition space would reflect a rapid, non-linear transition between action modes, as illustrated in Figure 1 (*left*), such that the transition area would, for example, be flanked by ~100% actualization in the top left area of the first action mode (e.g. grasping with one hand), and the second action mode seen bottom right (e.g. grasping with two hands). A particularly interesting characteristic of the transition-space is that both action modes are available (the state space is bi-stable). Thus, despite it being a fairly simple space, it reflects a wider set of possible transition dynamics. For example, as can be seen in Figure 1 (*right*), with regard to the plank-moving task investigated by Richardson and colleagues, individual participants could have (A) transitioned from either one-handed grasping to two-handed grasping in the center of the transition space, i.e., exhibited a non-linear transition between one- and two-hand grasping, (B) actualized one-handed grasping throughout, (C) transitioned to two-handed grasping before entering the transition space, or finally, (D) participants could have switched back and forth within the space between the different action modes.

It should be noted that the four different transition dynamics depicted in Figure 1 represent the set of possible action dynamics that could be observed (more or less) within a simple dichotomous action space (i.e., where the action modes are orthogonal and categorically distinct). In most real-life situations however, the same “functional” affordance can be actualized in many different ways. For example, if an organism would like to climb a riser that is too high to simply step up on, they would likely explore (and discover) multiple other ways to climb it. If the riser or surface is at chest height for example, depending on ability they could grab onto the edge of the surface and pull themselves up, or jump and pull themselves up, and so on. That is, in a riser-climbing task the organism-environment relationship entails a complex range or manifold of actualization possibilities. Thus, the many different ways in which individuals are able to actualize a given affordance or action possibility are perhaps best conceptualized as lying on an action-space manifold.

It is perhaps self-evident that investigating the degree to which action-space manifolds are able to capture the various potential action dynamics that can occur during affordance actualization requires employing experimental settings that offer a range of action possibilities. In other words, the task needs to include as few constraints as possible to allow for as many behavioral modes, action dynamics, and action-mode transitions as possible to emerge. Standing experimental methodologies, however, are typically highly constrained in order to isolate the behavioral mode of interest (Cartwright, 1983; 1999; Järvilehto, 1998; Bechtel & Richardson, 2010). For example, general instructions for a climb-ability study might take the following form: “*from a standing position judge if you could step onto the riser in front of you by lifting your foot up in front of the riser, placing your foot on it and*

heaving your body up with your leg”. These kinds of experimental restrictions on the ways an affordance can be actualized need to be reduced if one wishes to examine action-space manifolds. Experimentally, this essentially requires developing an *under-constrained* task context that allows for complex (multi-stable) organism-environment action dynamics to emerge.

In order to develop this kind of task methodology, Fajen’s (2007) conceptualization of soft and hard constraints provides an instructive guide. To define the role that soft and hard constraints have on the actualization of an affordance, imagine decelerating your car at a red light with another car stopping in front of you. To be successful in this task context (i.e., not hitting the car in front of you), you must brake enough to maintain (at least) a minimal space between your car and the car in front of you. In this case, the minimum amount of braking to succeed in the task illustrates the *hard constraint boundary*, meaning that crossing this boundary ensures task failure, i.e. a collision. However, assuming that one stays some distance away from this hard constraint boundary, that still leaves many different ways that braking can be implemented successfully. For example, one could brake heavily at the outset and keep a constant large distance. Alternatively, the complete opposite, riding bumper to bumper, is also a possibility. Presumably, there are many different factors that could influence braking behavior in such an under-constrained task context, continuously and dynamically pushing one’s position within the action space toward or away from the hard boundary. These factors are the *soft constraints* of the system and could e.g. be mood, goals, habits, ability, distractions, need for safety, and so on. In accordance with a dynamic systems perspective, the existence, relevance, and influence of different soft constraints are dependent on task context. Accordingly, understanding how and why people behave in a certain way in under-constrained situations is a function of both the hard constraints, defined by the task dynamics, and the soft constraints imposed on the task dynamics by the individual, environmental, and cultural-social factors. The current study was designed to investigate the emergence of varied behavioral dynamics characterizing behavioral (re-)organization within such an under-constrained task context.

Current Study

To understand why a varied range of action dynamics might emerge in a given affordance task context, it is necessary to consider what the underlying action space or manifold might look like. Although it is possible that different action dynamics might arise from different behavioral action spaces, of particular interest here was the degree to which a two-parameter state space defined by the Cusp Catastrophe system (Thom, 1975) could be employed to capture the varied action dynamics observed within an under-constrained affordance-task context. This system takes the form

$$\dot{x} = -x^3 + a + bx \quad (1)$$

where x is the state variable, a and b are system parameters, and the fixed points (or solutions; values x^* , which result in $\dot{x} = 0$) of the system for a range of a and b parameter values forms a two-dimensional manifold (depicted in Figure 2). The key feature of this

systems is that it entails both mono- and bi-stable dynamics depending on the value of the splitting factor b . Consequently, it has also been employed to capture various bifurcation phenomena, including human anxiety and performance, organizational order, developmental transitions, problem-solving and decision-making (e.g., Guastello, 1995; Hardy 1996; van der Maas & Hopkins, 1998; van der Maas & Jansen, 2003; Richardson, Dale & Marsh, 2014), as well as dating (Tesser, 1980), reaching and grasping (Wimmer, Savelsbergh, van der Kamp, & Hartelman, 1998), binge drinking, cognitive workload, and fatigue (Guastello, Malon, Timm, Weinberger, Gorin, Fabisch & Poston, 2014, for which it was also used as a statistical model).

A ball transportation task was devised where balls were presented to participants at the starting area in a sequential manner at increasing and decreasing intervals (14 to 2 seconds and vice versa), with interval corresponding to control parameter a . Participants were required to pick up the balls from the starting area and transport them to a box 9 meters away. Importantly, participants were free to complete this ball transportation task in any way they chose and no rewards or punishments were given to maintain the least amount of constraint possible (i.e., to promote the emergence of variable action dynamics). For each ball, distance moved before releasing the ball was measured and represents the order parameter (or state variable, x) for the system. This measure was chosen as it captured the most critical behavioral point of task performance, indicating the state of the system for each ball transportation event. Participants were expected to self-impose different constraints to varying degrees, e.g. motivation, intention, perceived ability, learned helplessness, etc. In the present experiment they were considered collectively as an uncontrolled time-varying parameter by means of the splitting factor, b .

With regard to the Cusp fixed point manifold illustrated in Figure 2, there are two general types of action (sequence) dynamics, *behavioral action series*, that are possible as interval (again, represented by the parameter a) is increased or decreased: *continuous* and *discontinuous* (where each point in an action series correspond to a single ball transportation event). Continuous series result when $b < 0$, with the series gradually increasing or decreasing as the value of a (time interval) is increased or decreased, respectively. Continuous series are also predicted when $b > 3$, but in this case correspond to steady state action series as a is scaled up or down¹. Importantly, the task context implies two qualitatively different steady state action series depending on the initial value of x (when scaling between -2 and 2 for a , while $b > 3$), with $x_{initial} > 0$, corresponding to a participant always running most of the way up to the box, whereas $x_{initial} < 0$ corresponds to a participant always throwing the ball from the start location. Finally, discontinuous series are predicted for $0 < b < 3$, when $-2 < a < 2$. The Cusp surface therefore specifies *three* specific types of action dynamics: (1) gradual increase or decrease in distance moved; (2) fixed (steady state) behavior (i.e., always moving the same distance irrespective of temporal interval); and (3) an abrupt or non-linear transition in action (i.e., distance moved).

¹Technically, when $b > 3$ series will exhibit a small gradual increase or decrease toward the end of each series of values. In the context of the task, however, this increase or decrease in either end of a series is not considered large enough to qualitatively change the type of behavioral dynamic observed.

Experiment 1 was designed to test whether a behavioral state space defined by the Cusp Catastrophe Model could be used to simulate action series data accounting for the expected wide range of behavioral variability observed between (and within) participants. In accordance with Chemero (2009) and Stepp, Chemero, and Turvey (2011), the model also served to guide further experimentation. Finding experimental confirmation of predictions derived from the model serves to improve confidence in the model. Two follow-up experiments were conducted where additional hard (Experiment 2) and soft (Experiment 3) constraints were imposed on participants. The hard constraint employed in Exp. 2 consisted of restarting the experiment in its entirety if a ball was presented when there already was another ball in the starting area, i.e. if balls began to stack up. There are, of course, other ways of implementing a hard constraint boundary, this particular one was chosen as it clearly distinguishes between success and failure (and reward/punishment). The soft constraint employed in Exp 3. emphasized accuracy through the addition of a points system, where each ball in the box counted for one point each and was tallied at the end of each action (ball presentation) series. The latter was considered a soft constraint as no additional consequences (punishments or rewards) including distinct success/failure criteria were imposed. Out of the potentially unlimited soft constraint candidates, this specific one was chosen as it highlighted an aspect already inherent in the task as opposed to adding a new one. We expected that the imposition of these hard and soft constraints would predictably reduce the ways in which participants behaved, in terms of both between and within participant action variability. Furthermore, corresponding restrictions on the parameterization of b in the Cusp Catastrophe Model were expected to account for the observed reduced variability.

2. Experiment 1

The task consisted of transporting balls, one at a time, from a starting area to a wooden box. To minimize the imposition of task constraints participants were free to choose how to act to complete the task (i.e., walk, run, throw, or any combination thereof from any distance away from the box). Participants completed two trial series, with each series including the sequential presentation of fifty-two balls. Half of the participants started their first series at an increasing rate: beginning with 14 second interval ball presentations, this interval was decreased incrementally by 1 second every four balls down to 2 seconds (i.e., four balls were presented one by one at each time interval). A break was offered after the first series and then the second series was started with the control parameter scaled in the reverse direction (i.e., from 2 to 14 seconds). The other half of participants completed these same two sequences in the reverse order (i.e., 2 to 14 second sequence, followed by the 14 to 2 second sequence). The first aim of Experiment 1 was to demonstrate the variability of behavioral dynamics that emerge from the underlying under-constrained task context. The second aim was to model and simulate this variability using the Cusp Catastrophe Model, in order to formally illustrate how self-imposed soft constraints operate to continuously modulate task performance.

2.1 Method

2.1.1 Participants—Sixty-nine undergraduates from the University of Cincinnati participated in the experiment for partial course credit.

2.1.2 Materials—At the ball pick-up location, the starting area (marked '2' in Figure 3), plastic playpen balls (17.76cm in diameter) were put through an angled PVC pipe (marked '4'). The mouth of the PVC pipe extended back through an opaque curtain hung from the ceiling (marked '3'). The end of the PVC pipe protruded into the starting area located on a wooden table (dimensions: 40cm wide, and 26.5cm deep). A large wooden box (marked '1', dimensions: 110cm wide, 55cm deep, and 120cm high) was positioned at nine meters from the back edge of the starting area. A computer program was used to visually signal an experimenter ('E') positioned behind the opaque curtain when to release the balls one by one. A video-camera ('V') was used to record participants' ('P') movements and actions throughout the experiment.

2.1.3 Task and procedure—Participants were told that the task involved transporting plastic playpen balls from a starting area to a wooden box located on the other side of the room. They could use one ball at a time and were instructed to get the balls into the wooden box, while at the same time avoiding a pile of balls to stack up in the starting area. They were told that the time between ball presentations would change from fast to slow or slow to fast (depending on sequence condition). They were also instructed that if they dropped a ball accidentally then it could be picked up; however, if they attempted to get it into the box but missed, then they should not retrieve it and instead move on to the next ball. Finally, they were instructed to solve the task in any way they liked. (There was no punishment if any rule was broken or for unsuccessful performance, and no incentives were given for accurate or fast performance).

2.1.4 Data analysis and behavioral classification—The distance that participants moved (walking or running) prior to releasing (throwing or putting) the ball was determined from line-markings on the floor, along with task success (i.e., whether participants successfully got the ball into the box or not), from the video recordings. The movement distances were analyzed using Matlab 2016a (MathWorks, MA), with the behavior exhibited by participants in each temporal series (i.e., 2 to 14 second and 14 to 2 second series) graphically classified into one of four categories of general behavioral dynamics (see below for more details).

Prior to classification, the movement distances were averaged over each change in time interval, i.e. the average distance moved prior to releasing the ball was calculated over the four balls that had a fourteen second interval, then the average distance moved was calculated over the four balls within thirteen second intervals, etc. This resulted in thirteen averaged movement distances for each 52-ball sequence (example action series of each type of dynamic are provided in Figure 5 for both increasing and decreasing interval sequences). From these behavioral action series two descriptive statistics were calculated: (1) mean distance moved (Dm); and (2) mean difference moved before versus after the maximum change in distance moved occurred (ΔDm). Dm was used for its ability to simply separate

between generally large or small distance moved. ΔDm was chosen based on its ability to indicate what kind of general dynamic is seen based on the type of variability within an action series. E.g. for a participant running all the way up to the box for almost all balls, but decides to throw a ball or two, it would be a mistake to characterize the entire action series as a non-linear dynamic. Using ΔDm correctly shows that although there is a large difference between singular values, there is still a small difference between the first and second half of the series. In contrast, throwing from afar for half the series and then running all the way up to the box, ΔDm increases, clearly indicating a large difference before and after the largest difference between singular datapoints. The two descriptive statistics were used to classify each behavioral time series as follows:

- Stable (fixed) small distance moved (stD_{small}), whereby participants essentially always throw the ball from shorter than half of the distance to the bin. More specifically, $Dm < 4.8$ meters and $\Delta Dm < 1.58$ meters.
- Stable (fixed) large distance moved (stD_{large}), whereby participants essentially always moved longer than half of the distance to the bin prior to releasing the ball. More specifically, $Dm > 4.8$ meters and $\Delta Dm < 1.58$ meters.
- Gradual change (phase transition) in distance moved ($ptD_{gradual}$), whereby participants gradually increased or decreased the distance as time interval decreased or increased, respectively (i.e., an inverse relationship between distance moved and time interval). More specifically, $1.58 < \Delta Dm < 3.8$ meters.
- Non-linear change (phase transition) in distance moved ($ptD_{nonlinear}$), whereby participants exhibited a large or non-linear change in distance moved across a small change in time interval. More specifically, $\Delta Dm > 3.8$ meters.

The demarcation values were determined using a 10-fold, J48 decision tree classifier in WEKA 3.8 (Witten, Frank, Hall & Pal, 2016), correctly classifying 99.28% of the dynamics. The specific values determined by the classifier varied trivially due to thin corridors of uninhabited space between the categories. It should be noted that the categories only serve to clarify the general differences between different types of behavioral dynamics in the context of this particular task. Therefore, the similarity of a pair of ΔDm and Dm values for two dynamics are more telling of how similar two specific dynamics are, regardless of if the two happen to fall on either side of a demarcation line. For our purposes here, however, the collection of solutions and the space they occupy is more important than detailed comparisons between individual action series.

2.1.5 Modeling and simulation—Traditionally, the b -parameter in the Cusp Catastrophe Model is treated as a static variable, fixing it at a specific value and then scaling a up or down. In fact, this is how the exemplar action series plotted on the solution surface in Figure 4 (*left*) were generated (i.e., by fixing the value of b and then scaling a for a given initial value x_0). With regard to the current task, this would be equivalent to assuming that nothing would have an impact on behavior over the action series. However, in the present experiment b stands in for the collection of self-imposed soft constraints, which varied both during and across the action series. For instance, an individual's motivation or goal intention may have been continuously modulated during the task. Thus, at each interval change (or

ball for that matter), the resulting distance moved may reflect a continuous (or discrete) change in b . Consequently, although the initial value of b might have been 1, the action series does not necessarily have to take a straight path resulting in a non-linear transition. In fact, it could wander in any direction depending on how the constraints modulate ongoing behavior (i.e., the value of b is modulated) for a specific individual (see example action series in Figure 4 *right*). Accordingly, a range of action series were generated using the Cusp Catastrophe system by scaling a in interval steps consistent with the time interval steps employed in the experimental study (i.e., from 2 to -2 in 13 steps, equal to 14 to 2 seconds in Figure 4), as well as scaling the initial b -value from 3.5 to -2.5. Importantly, the value of b was also changed at each interval step, by Δb , which was calculated by drawing values from a random distribution ($M = -.61$, $SD = 1.65$ when scaling up, $M = .61$, $SD = 1.65$ when scaling down, see Figure 4 *right* for example action series). This produces semi-random walk where the next b value is conditionally dependent on the previous one. The initial value for x was set to 2 for descending simulations, and a uniformly distributed random number generator scaled to x (-2 to 2) was set for ascending simulations.

This difference in initial x value stems from underrepresentation of stD_{small} in the experimental data. The simulation produces two action series, one for each scaling direction, and was iterated 70 times to match participant data ($N = 69$ rounded to nearest ten). Simulated pairs of action series were rescaled by adding the minimum absolute value to the dataset and then multiplying them by the ratio between the maximum distance in the experimental data and the maximum value in the data points.

2.2 Results

As can be seen from an inspection of Figures 5 and 6, and Table 1, participants produced a wide range of different behavioral dynamics (including all four of the expected ones). The variability within and across human participants and ball sequences is most easily discerned from an inspection of Figure 6 (*top*), in which classification is plotted as a function of mean distance moved (Dm) and difference in mean distance pre-post maximum change in distance moved (ΔDm). The simulated action series also produced a comparable set of behavioral dynamics and classifications. A curve estimation analysis was conducted on the total frequencies of each distance across all data-points, revealing a linear increase in frequency as distance increased ($\beta = .84$, $t(33) = 8.81$, $p < .01$, where x = distance moved). A two-tailed, bivariate correlation analysis was run to investigate the relationship between distance moved and success (hit) versus failure (miss), revealing a positive association ($r = .64$, $p < .01$). Not surprisingly, as distance moved increased so did the probability of success.

2.3 Discussion

Experiment 1 investigated the variability in behavioral dynamics that occurred within an under-constrained ball transportation task. As can be seen from an inspection of Figure 6 (*top*) above, time interval operated as a control variable, pushing the system through different behavioral modes with a wide degree of variability both between and within participants. Indeed, the lack of definitive guidance on how to succeed (or fail) in the task appeared to induce participants to explore varying task solutions on their own. Simulations using the Cusp Catastrophe equation with a semi-random, time-varying b -parameter

captured the wide range of dynamics observed between and within participants. Note that this does not imply that human behavior changes in a random or uncontrolled manner. Here, random variation in b is merely standing in for fluctuations in undefined behavioral variables that vary both between and within participants dynamically. Of central importance, stable state linear, non-linear, and transient behavioral patterns all emerged from the same state space.

Simulations using the Cusp Catastrophe equation were conducted to test for deeper similarities than just the general shape of the four types of dynamics. Specifically, manipulating the b -parameter (adding a value from a random distribution to the previous b -value on each step across the surface) was key to explicitly demonstrating how the different behavioral dynamics could have emerged due to the under-constrained task context (see Figure 5 for comparisons between simulated and human data). Without manipulating the b -parameter dynamically, similar distributions can only be achieved by manually setting each b -parameter to correspond to participant data (but would defeat part of the purpose to simulate data). Even with this manipulation, however, individual action series would be too smooth to resemble human within-series variability. This variability is directly driven by the non-static b -parameter in the model and, jointly with the characteristics of the random distribution, creates a drift (or walk) over the solution surface. Drift can push the b -parameter from an initial value that would have resulted in a non-linear transition, for example, down to a b -value that results in a gradual one. The simulations conducted here therefore suggest that there was a continuous modulation of soft constraints during task performance.

The similitude between real and simulated action series also demonstrates how the different behavioral modes emerge as a function of the under-constrained task context. Furthermore, the emergence of stable state linear, non-linear, and transient behavioral patterns from the same underlying state space points to a need for identifying and defining the constraints in the system at hand. Two follow up experiments with additional constraints were conducted in order to determine what effect specific constraints had in terms of restricting or encouraging certain behavioral dynamics.

3. Experiment 2

In Experiment 2, a hard constraint was chosen to strictly limit specific dynamics with a clear rule that specified task failure. Using the solution surface of the Cusp Catastrophe Model as a guide, we chose a constraint that prohibit any solutions from the top part of the bi-stable region on the Cusp surface (highlighted region marked '1' in Figure 9). This was instantiated in the task as a rule: if a ball is presented while there already is a ball in the starting area (i.e. if balls begin to stack up), participants were punished by restarting the entire experiment. Since the restricted surface contains higher values of x , corresponding to longer distances in the task, the expectation was that the rule would severely restrict stD_{large} solutions.

3.1 Method

3.1.1 Participants—Thirty University of Cincinnati undergraduates participated for partial course credit.

3.1.2 Materials, task, and procedure—The same materials used in Experiment 1 were employed for Experiment 2 except that the distance was shortened to 8 meters. The task and instructions also remained the same except for the addition of a hard constraint: only one ball was allowed in the starting area at any point in time. If a ball ever came out when there already was a ball in the starting area, the series was immediately ended and restarted.

3.2 Results

Participant distance measures were rescaled from 8 to 9 meters, and as can be seen in Figure 7 and Table 2, participants exhibited a wide range of dynamics except for *stD_{large}*. Simulations using a modified solution surface where stable fixed points (for $x > 0$) within the bistable subspace were restricted, produced similar distributions and behavioral dynamics compared to participant data (see Figure 7 *bottom*).

3.3 Discussion

Imposing a hard constraint that essentially limits participant behavior to guarding the starting area was predicted, and found, to reduce *stD_{large}* dynamics. Participants were effectively unable to stably produce movement over larger distances and, in terms of the model's solution surface, the action series that would have fallen on the top part of the cusp (see Figure 9, region '1') were untenable. The function of the hard constraint is clear: to be successful in the task certain behavioral dynamics were unavailable, in this particular version of the task those corresponding to *stD_{large}* dynamics.

4. Experiment 3

The way in which participants were punished if they moved too far from the starting area in Experiment 2 successfully restricted participants from maintaining moving a large distance. In Experiment 3, the imposed constraint aimed at encouraging accuracy, without directly restricting any action dynamics. This was instantiated by adding a points system, where the number of balls in the box were counted after each block of trials and participants were verbally encouraged to beat their first score in the second series of trials. It was expected that this would push participants toward *stD_{large}* (and away from *stD_{small}*) action dynamics, and also in general that participants would use longer distances (in order to maximize accuracy).

4.1 Method

4.1.1 Participants—Thirty University of Cincinnati undergraduates participated for partial course credit.

4.1.2 Materials, task, and procedure—The same materials used in Experiment 2 were employed for Experiment 3. The task and instructions also remained the same except exchanging the hard constraint for a soft one: emphasizing accuracy. Participants were encouraged to get balls into the box by the addition of a points system, where 1 point was given for each ball that was thrown, or put, into the box. Accuracy was further emphasized by verbally counting the total number after each series by the experimenter, and by being told that the goal was to increase their score in the second series.

4.2 Results

As can be seen in Figure 8 and Table 2, participants again exhibited a wide range of behavioral dynamics, although there was a marked increase in stD_{large} dynamics and the stD_{small} dynamics are all close to the stD_{large} and $ptD_{gradual}$ borders. Simulations were conducted in a similar manner to Experiment 2 except for sampling initial values of b from a smaller set ($3.5 < b < 3$) when scaling up (i.e. when initial $a = -2$).

4.3 Discussion

Because accuracy decreases with decreases in movement distance, it was expected and found that encouraging accuracy would increase movement distances and therefore incur a decrease in stD_{small} dynamics. In terms of the Cusp Catastrophe Model, action series are pushed away from the lower part of the Cusp surface (see Figure 9, region '2'). Figure 8 shows clear avoidance of moving short distances. Although there are a couple of action series that still fall within the pre-defined limits for stD_{small} action dynamics, it is important to note that two action series close together in Figure 8 (or Figure 6 and 7) but that fall on either side of a demarcation line are not considered fundamentally different. Rather, the underlying space is considered continuous, as the Cusp Catastrophe Model's solution surface itself implies.

5. General Discussion

The current study was designed to investigate how task constraints influence the variance of action dynamics during affordance actualization. Participants completed a ball-transportation task, where balls were presented in a starting area at an interval that first decreased (from 12s to 2s) and after a short break increased (or vice versa). As expected, the results of Experiment 1 revealed large between- and within-participant variability with regards to the action dynamics participants exhibited, an emergent consequence of allowing participants to self-impose constraints throughout task performance. Despite the varied results, the Cusp Catastrophe Model successfully captured the behavioral data, illustrating how the varied dynamics emerged when varying the splitting factor b . To further explore the mapping between parameter specific behavioral constraints and action variability, Experiment 2 imposed a hard constraint consisting of a rule that prevented participants from allowing balls to stack up. This predictably pushed participants away from action dynamics involving sustained longer distances (i.e., stD_{large}), as shorter intervals afford incrementally less distance traversed. Interestingly, highlighting interval as a key variable for success seemed to attract participants toward coordinating their movement distance to the interval, seen by the large increase in gradually increasing/decreasing action dynamics (i.e., $ptD_{gradual}$). Experiment 3 imposed a points system consisting of counting the number of balls that were in the box after each series and participants were verbally encouraged to increase their score in the second series. This soft constraint encouraged participants to traverse larger distances, almost eliminating action series involving sustained shorter distances (i.e., stD_{small}). Both follow-up experiments decreased the number of non-linear solutions, suggesting that imposing constraints within these task dynamics shifts stable behavioral modes toward, arguably simpler, linear ones.

As noted in the introduction, literature on affordance boundaries and behavioral mode switching has usually focused on pre-defined bi-stable state spaces. In the case of affordance boundaries, the point of transition is between being able to actualize an affordance and not being able to: e.g. being able to step up on a riser, or not (Warren, 1984). For behavioral mode transitions there has been a focus on switching between one distinct way of actualizing an affordance to another, e.g. one-handed to two-handed grasping, or two-handed grasping to grasping with one hand and an extending tool (Richardson et al., 2007). Misreading this literature (even slightly) may run the risk of understanding affordances as an either/or phenomenon, i.e. you either succeed or fail at actualizing an affordance, and the difference between one way of actualizing an affordance and another is clear and distinct. While the task in the current experiment contained these two distinct extremes (no movement and throwing compared to running/walking all the way up to the box and putting), participants were able to move as far, or short, into the space as they wished. This forms the basis for a continuous and gradual shift between the two distinct behavioral modes, resulting in a more complex, but arguably more realistic, perspective of affordance actualization. The simplified, categorical (and often dichotomous) distinction begins to fall apart when put in this perspective of smooth and continuous transfer between two (perhaps clearly distinct) behavioral modes. Additionally, in the present task the goal-criteria can seem clear cut: getting the ball in or not. This might seem to imply a second dichotomous relationship, namely between success and failure. However, in the control condition this dichotomy is questioned by the vagueness, or indifference, of the task goal. That is, participants may have self-imposed constraints depending on how they interpreted the instructions, but from the dynamics that emerged in the experiment it is clear that some were driven by performance and others not. For example, some focused on being 100% accurate by running, but with the consequence that they couldn't use all the balls when the interval between balls was short. Others also focused on accuracy but wanted the opportunity to use all balls and were forced to throw. Future research could delve deeper into specific reasoning and motivation by reviewing the video with participants and asking them to explain their behavior. Success or failure here is not clear cut, but rather, just like the characterization of affordances themselves, is more continuous and gradual than dichotomous. Ultimately, the point here is to theoretically and empirically clarify the continuous nature of affordance actualization and their fulfilment criteria.

The three experiments illustrate the direct correspondence between the mathematical model's state space, the availability of stable behavioral modes for participants, and the subsequent restrictions in availability of both of these due to the addition of constraints. The modeling and simulation of the under-constrained version of the task in Experiment 1 provided a better general understanding of what role parametric constraints play during the actualization of an affordance. Here, parameter a is already clearly defined as the control parameter (i.e., interval), but the b -parameter was intentionally left as under-constrained as possible. Experiment 2 and 3 therefore attempted to showcase two ways in which to further specify what constraints the b -parameter may map onto. One of the key features of the two follow-up experiments is the model-driven predictions about possible dynamics. By no means is it the intent of these simulations to produce identical layouts of dynamics (as in Figure 7 and Figure 8 above) or even identical distributions (as in Table 2). However, the

model should be able to produce a similar variety and spread of dynamics corresponding to the behavioral expectations of participants depending on the constraint at hand.

This study also speaks to a controversy about the value of dynamical modeling in the cognitive sciences. Several philosophers of science have suggested that dynamical models do not explain, but only describe, data (e.g., Kaplan and Bechtel 2011). The experiments here provide a case of a dynamical model going beyond merely describing experimental results. The model of the data from Experiment 1 does indeed describe the data gathered there. However, the Cusp Catastrophe Model was the inspiration for the manipulations in Experiment 2 and 3, and the results of these experiments together with the corresponding simulations confirm the appropriateness of the model. That is, the Cusp Catastrophe Model is not just a description of the data in this study, but is also serving as a guide to discovery, (Chemero 2009; Stepp et al., 2011) inspiring further experimentation and subsequent modeling.

It should be noted that two important aspects of the Cusp Catastrophe (or indeed any state space with a bi-stable region) have not been explicated in the present research, limiting conclusions about within-series dynamics. If b is held constant at a value of e.g. 1 and an action series is simulated across and back over the Cusp solution surface, the transition point from the top to the bottom part of the cusp will be different depending on the direction of the action series. Essentially, the system will remain in its initial state for longer before it transitions compared to if it begins in the other direction (in which case it will remain longer in that initial state). This is termed ‘hysteresis’ and indicates that the current state of the system is dependent on its history (Kelso, 1984; 1995; also see Lopresti-Goodman, Turvey, & Frank, 2013; for negative hysteresis). The second aspect concerns signs indicating that a non-linear transition might occur, e.g. critical slowing down and critical fluctuations (Kelso, 1995). The former refers to increased variability in dynamics observed prior to a transition and the latter refers to increased time-to-recovery if the system is perturbed as it approaches a transition (Haken, 1977; also see Gilmore, 1981; and van der Maas & Hopkins, 1998; for further signs, or catastrophe flags). It would be interesting for future research to focus on these signs (or flags), as specifying what they constitute in any given context could be of great predictive value.

Generally, the findings speak to a methodological argument for clear identification and definition of the constraints actively shaping ongoing human behavior in any given task. The methodology reported here could also open up for an additional way to construct experiments. Instead of constraining behavior down to one (often highly specific) way of actualizing an action possibility, it is possible to let stable states naturally emerge from an underlying, under-constrained system. In this way, one can fully explore which solutions (and dynamics) are available (or common) in a given task or situation *first*, before restricting them by imposing constraints in an experiment’s other conditions. This sequence of events produces an overview of all the ways in which a task can be solved and then how a manipulation restricts those solutions. As it stands, constrained control conditions imply one specific solution, not necessarily the “normal”, general, or common one. Therefore, it will always rely on the specific project whether or not this condition is representative of the constructs that they attempt to capture. It should be conceded however that some phenomena

are difficult to operationally define, in which case highly constrained control conditions can be appropriate. Another future application of the present methodology would be to navigate unexplored phenomena, to uncover stable behavioral modes where a priori identification may be difficult (or indeed impossible). For example, traditionally difficult constructs to measure, like creativity and imagination, could potentially be explored effectively within this framework.

Acknowledgments

This research was supported, in part, by the National Institutes of Health (R01GM105045) and the Charles Phelps Taft Research Center. We are also grateful for the reviewers' insightful comments and suggestions.

References

- Bechtel W, & Richardson RC (2010). *Discovering complexity: Decomposition and localization as strategies in scientific research*. Cambridge, MA: MIT Press.
- Carello C, Groszofsky A, Reichel FD, & Solomon HY (1989). Visually perceiving what is reachable. *Ecological Psychology*, 1, 27–54.
- Cartwright N (1983). *How the laws of physics lie*. Oxford, UK: Clarendon Press.
- Cartwright N (1999). *The dappled world*. Cambridge, UK: Cambridge University Press.
- Cesari P, & Newell KM (1999). The scaling of human grip configurations. *Journal of Experimental Psychology: Human Perception and Performance*, 25, 927–935. [PubMed: 10464939]
- Chemero A (2009). *Radical embodied cognitive science*. Cambridge, MA: MIT Press.
- Fajen BR (2007). Affordance-based control of visually guided action. *Ecological Psychology*, 19, 383–410.
- Fitzpatrick P, Carello C, Schmidt RC, & Corey D (1994). Haptic and visual perception of an affordance for upright posture. *Ecological Psychology*, 6, 265–287.
- Gibson JJ (1979). *The ecological approach to visual perception*. Boston, MA: Houghton-Mifflin.
- Gilmore R (1981). *Catastrophe theory for scientists and engineers*. New York: Wiley.
- Guastello SJ (1995). *Chaos, catastrophe, and human affairs: Applications of nonlinear dynamics to work, organizations, and social evolution*. Mahwah, NJ: Lawrence Erlbaum.
- Guastello SJ, Malon M, Timm P, Weinberger K, Gorin H, Fabisch M, & Poston K (2014). Catastrophe models for cognitive workload and fatigue in a vigilance dual task. *Human Factors*, 56, 737–751. [PubMed: 25029898]
- Haken H (1977). *Synergetics: An introduction*. Heidelberg: Springer-Verlag.
- Hardy L (1996). Testing the predictions of the cusp catastrophe model of anxiety and performance. *The Sport Psychologist*, 10, 140–156.
- Hristovski R, Davids K, & Araújo D (2006). Affordance-controlled bifurcations of action patterns in martial arts. *Nonlinear Dynamics, Psychology, and Life Sciences*, 10, 409–444.
- Järvilehto T (1998). The theory of the organism-environment system: I. Description of the theory. *Integrative physiological and behavioral science*, 33, 321–334. [PubMed: 10333975]
- van der Kamp J, Savelsbergh GJP, & Davis WE (1998). Body-scaled ratio as a control parameter for prehension in 5-to-9-year-old-children. *Developmental Psychobiology*, 33, 351–361. [PubMed: 9846238]
- Kaplan D, & Bechtel W. (2011). Dynamical models: An alternative or complement to mechanistic explanations? *Topics in Cognitive Science*, 3, 438–44. [PubMed: 25164303]
- Kelso JAS (1984). Phase transitions and critical behavior in human bimanual coordination. *American Journal of Physiology: Regulatory, Integrative, and Comparative Physiology*, 15, R1000–R1004.
- Kelso JAS (1995). *Dynamic patterns: The self-organization of brain and behavior*. London: MIT Press.
- Lopresti-Goodman SM, Turvey MT, & Frank TD (2013). Negative hysteresis in the behavioral dynamics of the affordance “graspable”. *Attention, Perception, & Psychophysics*, 75, 1075–1091.

- van der Maas HLJ, & Hopkins B (1998). Developmental transitions: So what's new? *British Journal of Developmental Psychology*, 16, 1–13.
- van der Maas HLJ, & Jansen BRJ (2003). What response times tell of children's behavior on the balance scale task. *Journal of Experimental Child Psychology*, 85, 141–177. [PubMed: 12799166]
- Mark LS (1987). Eyeheight-scaled information about affordances: A study of sitting and stair climbing. *Journal of Experimental Psychology: Human Perception and Performance*, 13, 361–370. [PubMed: 2958585]
- MathWorks. (2016). MATLAB and Statistics Toolbox Release 2016a. Natick, MA: The MathWorks Inc..
- Richardson MJ, Dale R, & Marsh KL (2014). Complex dynamical systems in social and personality psychology. *Handbook of Research Methods in Social and Personality Psychology*, 253.
- Richardson MJ, Marsh KL, & Baron RM (2007). Judging and actualizing intrapersonal and interpersonal affordances. *Journal of Experimental Psychology: Human Perception and Performance*, 33, 845–859. [PubMed: 17683232]
- Stepp N, Chemero A, & Turvey M (2011). Philosophy for the rest of cognitive science. *Topics in Cognitive Science*, 3, 425–37. [PubMed: 25164302]
- Tesser A (1980). When individual dispositions and social pressure conflict: A catastrophe. *Human Relations*, 33, 393–407.
- Thom R (1975). *Structural stability and morphogenesis*. Reading, MA: W. A. Benjamin.
- Warren WH (1984). Perceiving affordances: Visual guidance of stair climbing. *Journal of Experimental Psychology: Human Perception and Performance*, 10, 683–703. [PubMed: 6238127]
- Warren WH, & Whang S (1987). Visual guidance of walking through apertures: Body-scaled information for affordances. *Journal of Experimental Psychology: Human Perception and Performance*, 13, 371–383. [PubMed: 2958586]
- Wimmer RH, Savelsbergh GJP, van der Kamp J, & Hartelman P (1998). A developmental transition in prehension modeled as a cusp catastrophe. *Developmental Psychobiology*, 32, 23–35. [PubMed: 9452905]
- Witten IH, Frank E, Hall MA, & Pal CJ (2016). *Data mining: Practical machine learning tools and techniques*. Burlington, MA: Morgan Kaufmann.

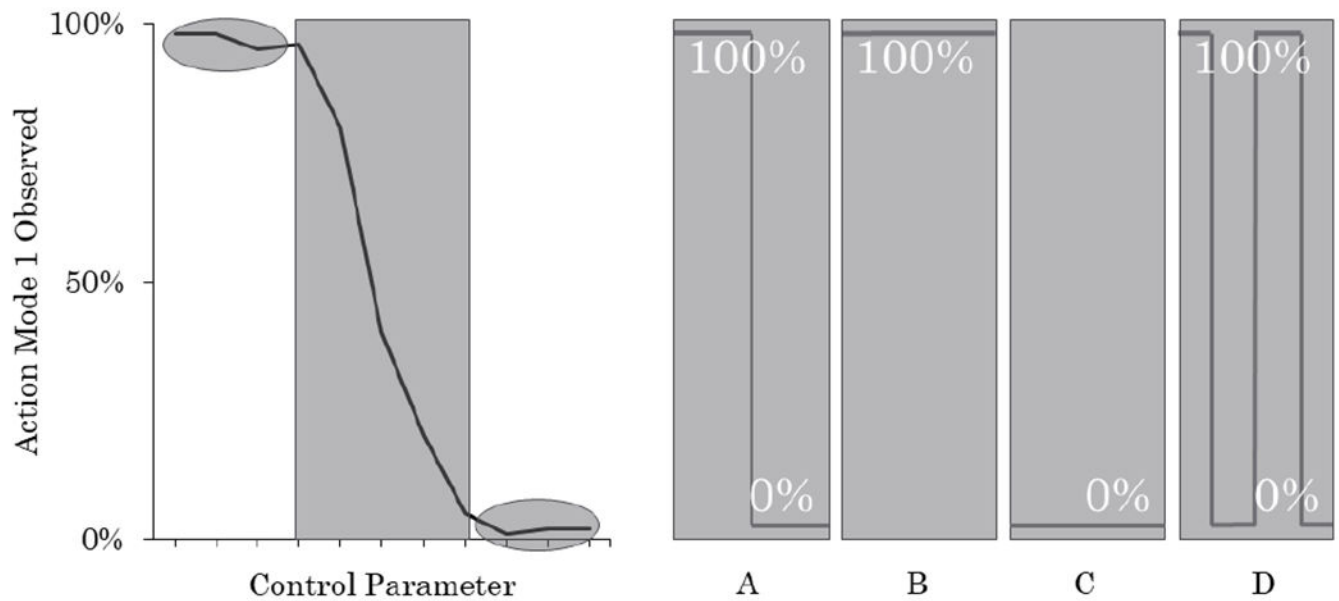


Figure 1.

(*left*) An illustration of group average actualization probability of bi-stable action modes at different levels of a control parameter. Top left area indicates ~100% actualization of Action Mode 1, middle area shows a non-linear transition between Action Mode 1 and 2, and bottom right area indicates ~100% actualization of Action Mode 2. (*right*) Four individual non-linear and linear types of dynamics from one action mode to another with A depicting a non-linear transition in the center of the subspace, B and C depicting maintenance of an action mode, and D depicting multiple switches between action modes.

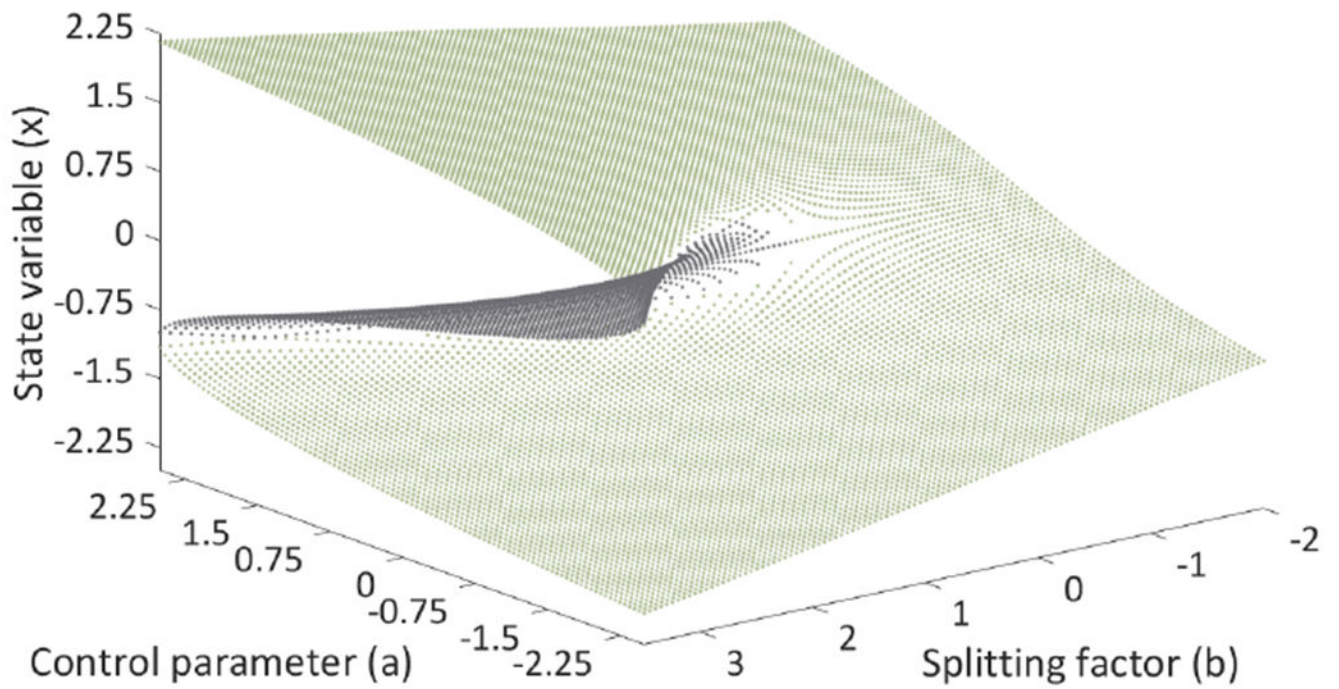


Figure 2. Cusp Catastrophe fixed point manifold. Each point on the manifold corresponds to the stable (grey) or unstable (black) fixed points, x^* , for a given parameterization of the control parameter, a , and the splitting factor, b .

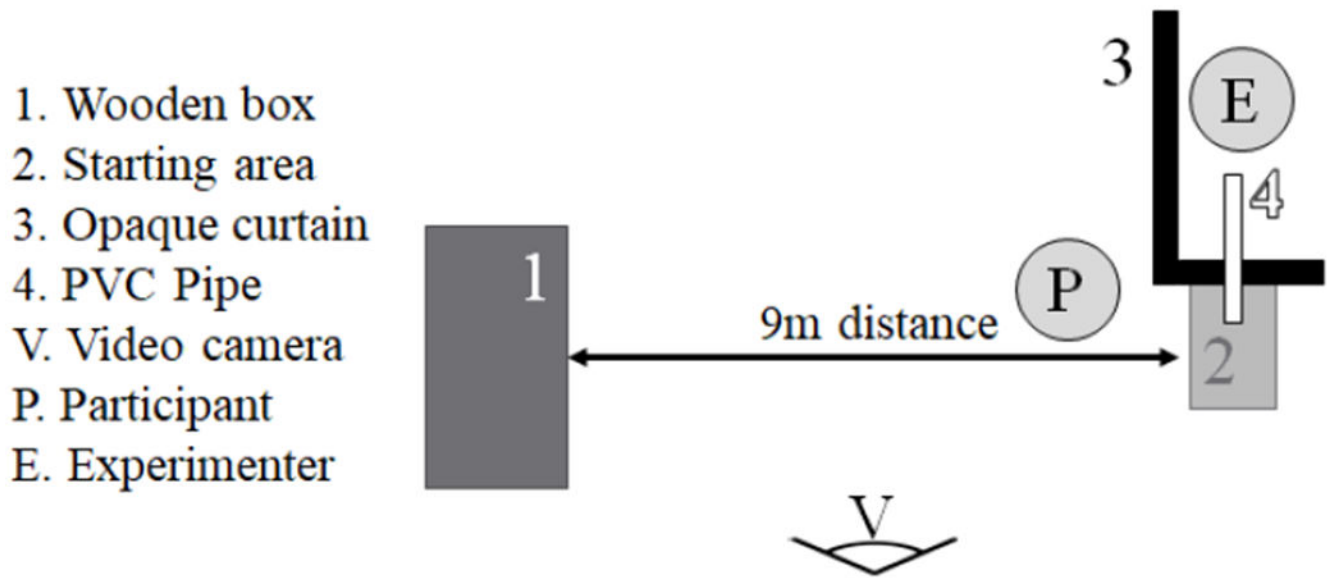


Figure 3.
General experimental setup.

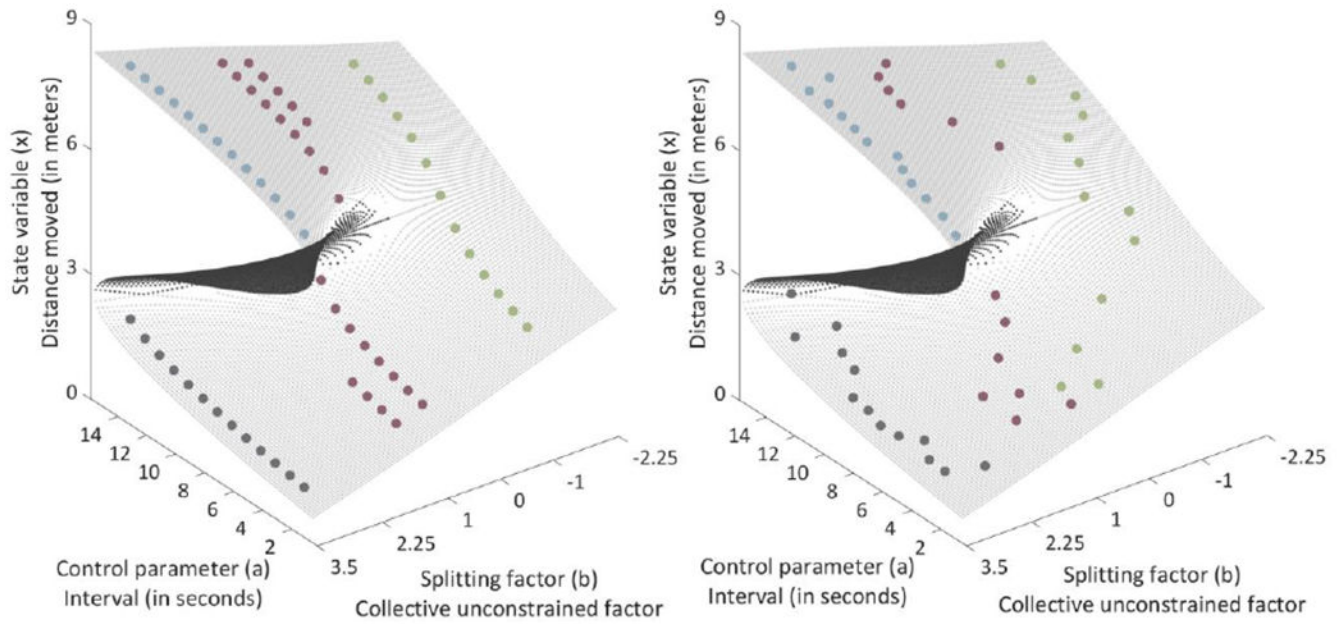


Figure 4. Cusp Catastrophe Model solution surfaces with action series simulated with static (*left*) and variable (*right*) b -parameters. Green points represent exemplar $ptD_{gradual}$ behavioral trajectories. Red points represent exemplar $ptD_{nonlinear}$ behavioral trajectories. The blue and dark gray points represent exemplar stD_{large} and stD_{small} trajectories, respectively.

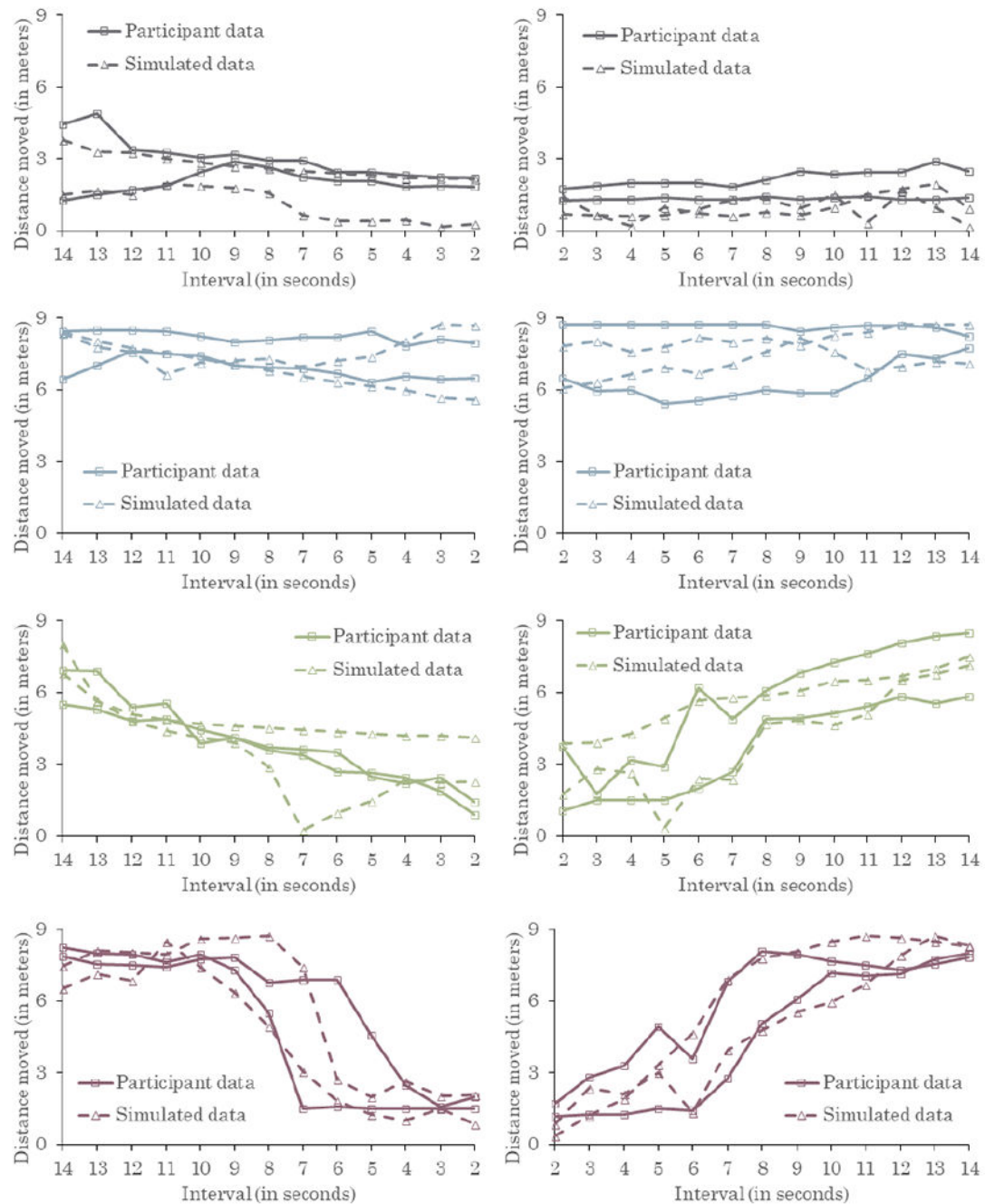


Figure 5.

Two examples each of participant (full line, square markers) and simulated (dotted line, triangle markers) action series: *stDsmall* (top), *stDlarge* (second), *ptDgradual* (third) and *ptDnonlinear* (bottom).

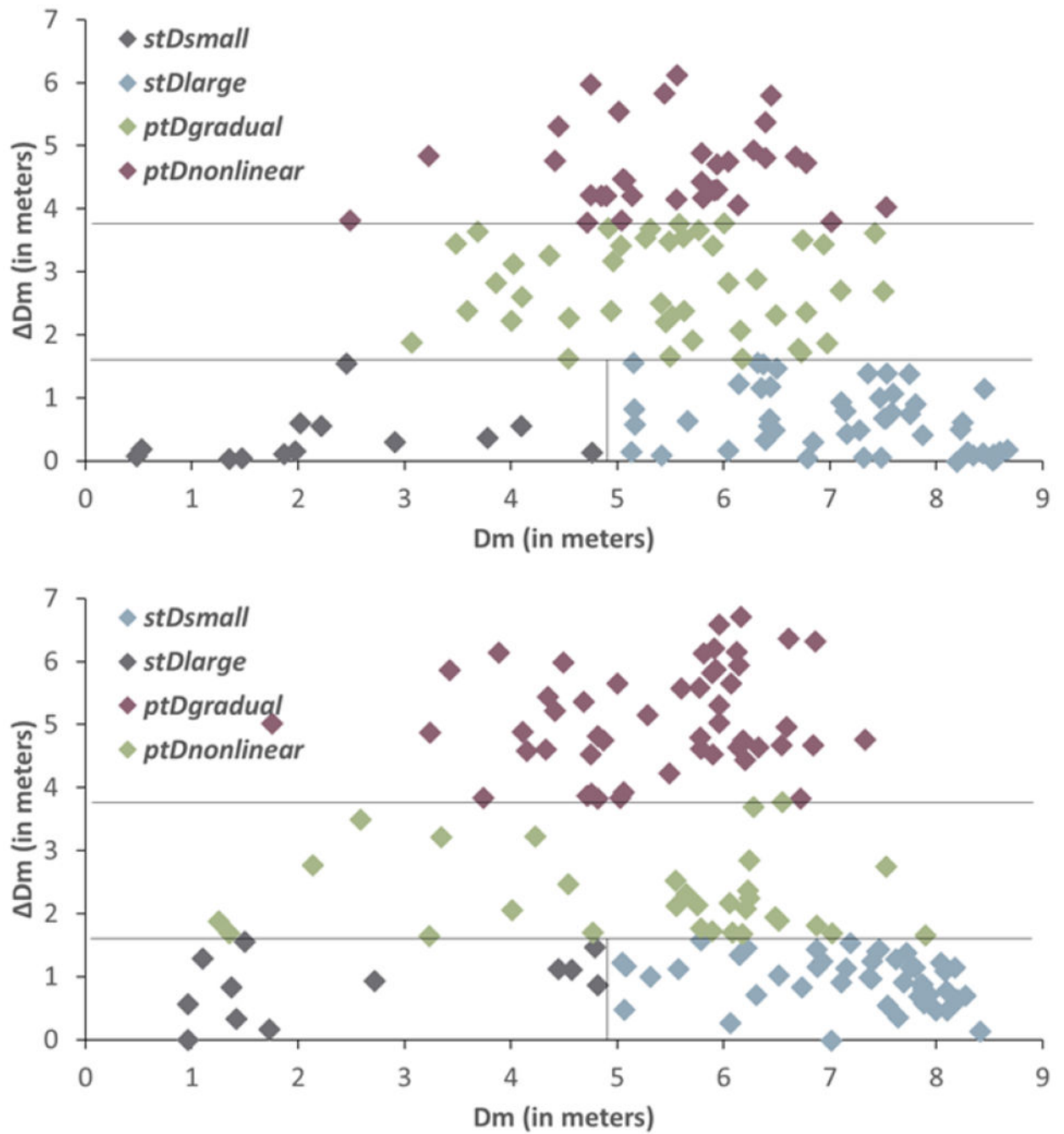


Figure 6. Participant (*top*) and simulated data (*bottom*) behaviorally classified as a function of mean distance moved (Dm) and difference in mean pre-post max change in distance (ΔDm).

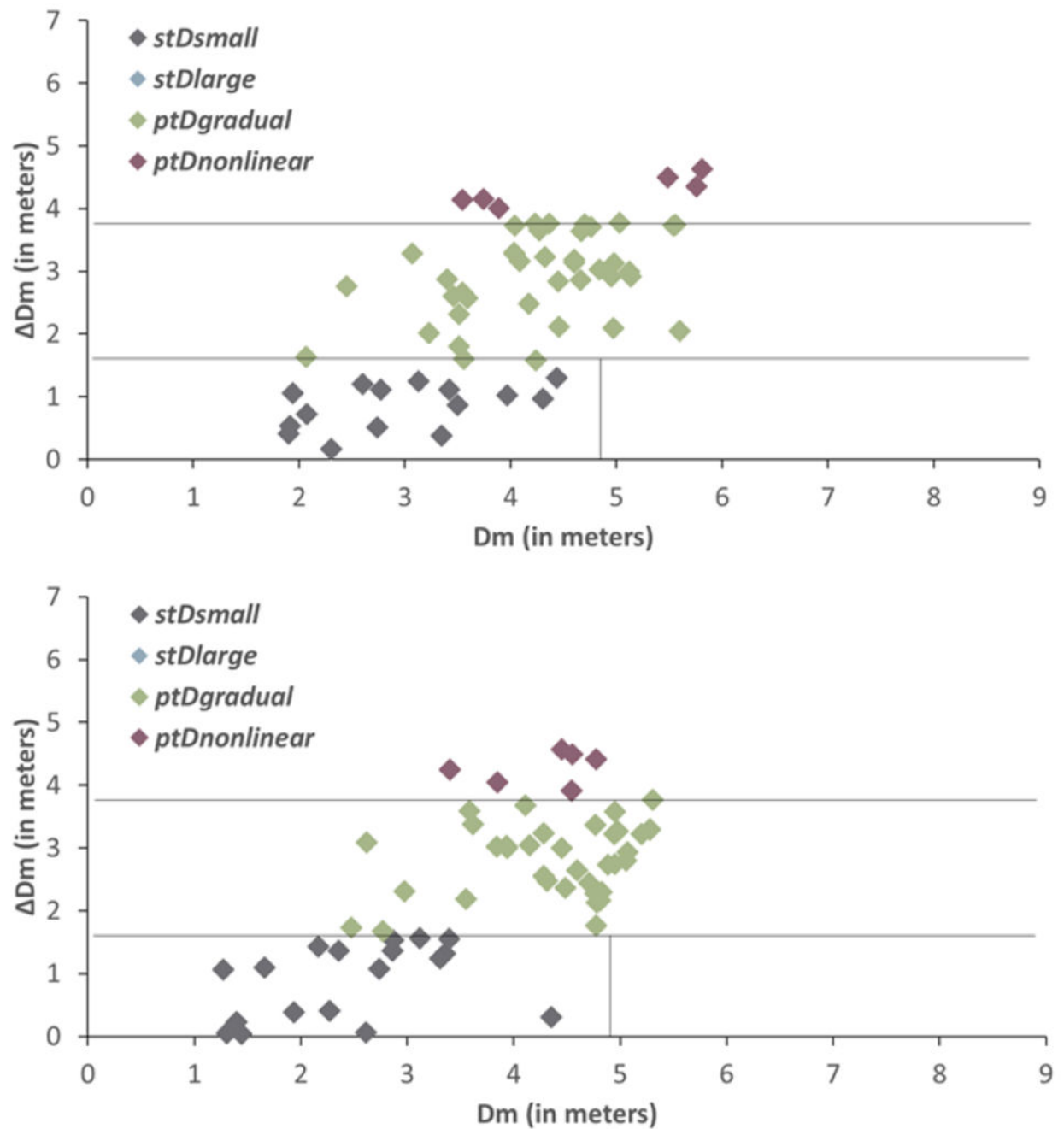


Figure 7.

Participant (*top*) and simulated data (*bottom*) behaviorally classified as a function of mean distance moved (Dm) and difference in mean pre-post max change in distance (ΔDm).

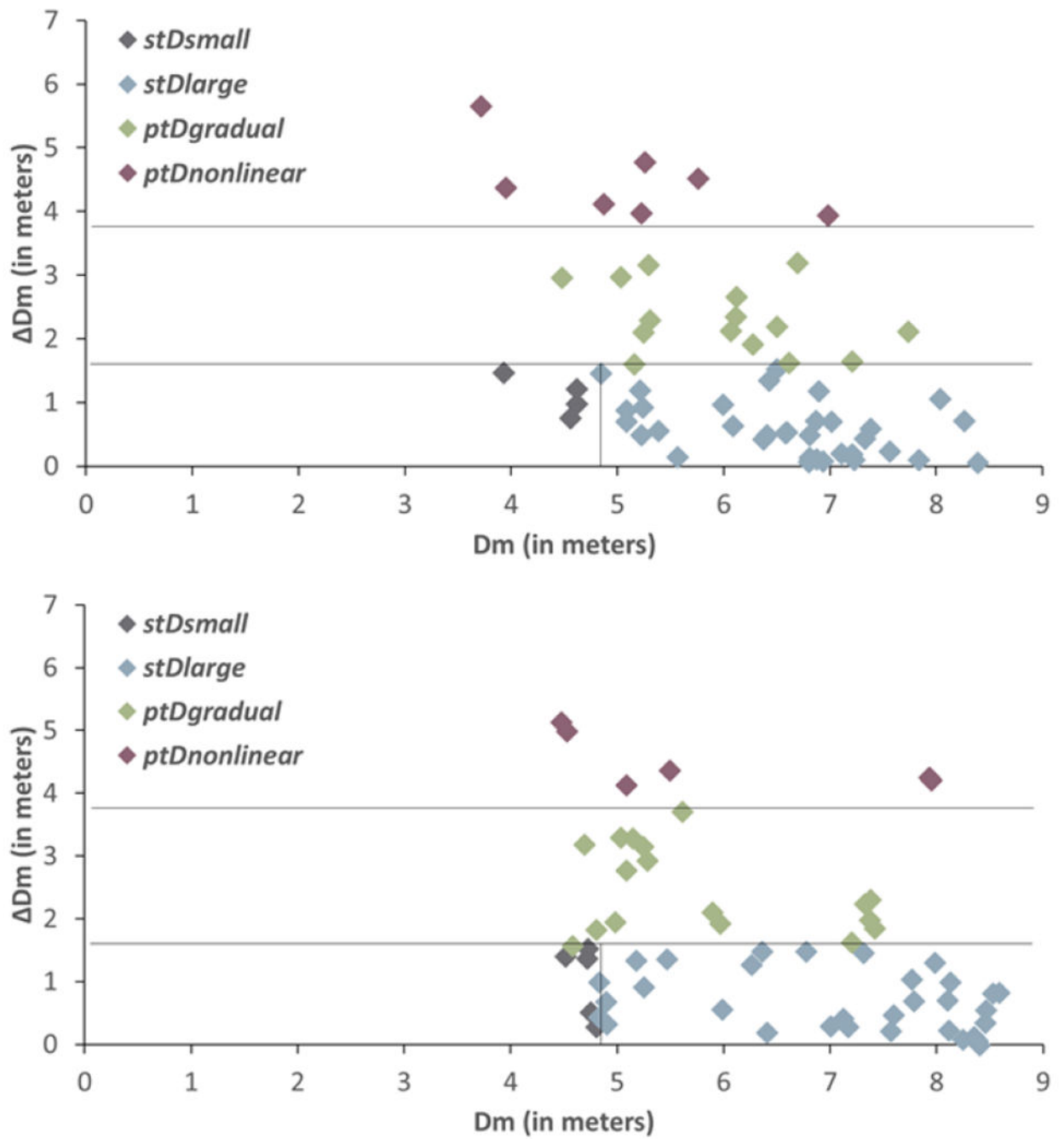


Figure 8. Participant (*top*) and simulated (*bottom*) behavioral classification as a function of mean distance moved (Dm) and difference in mean pre-post max change in distance (ΔDm).

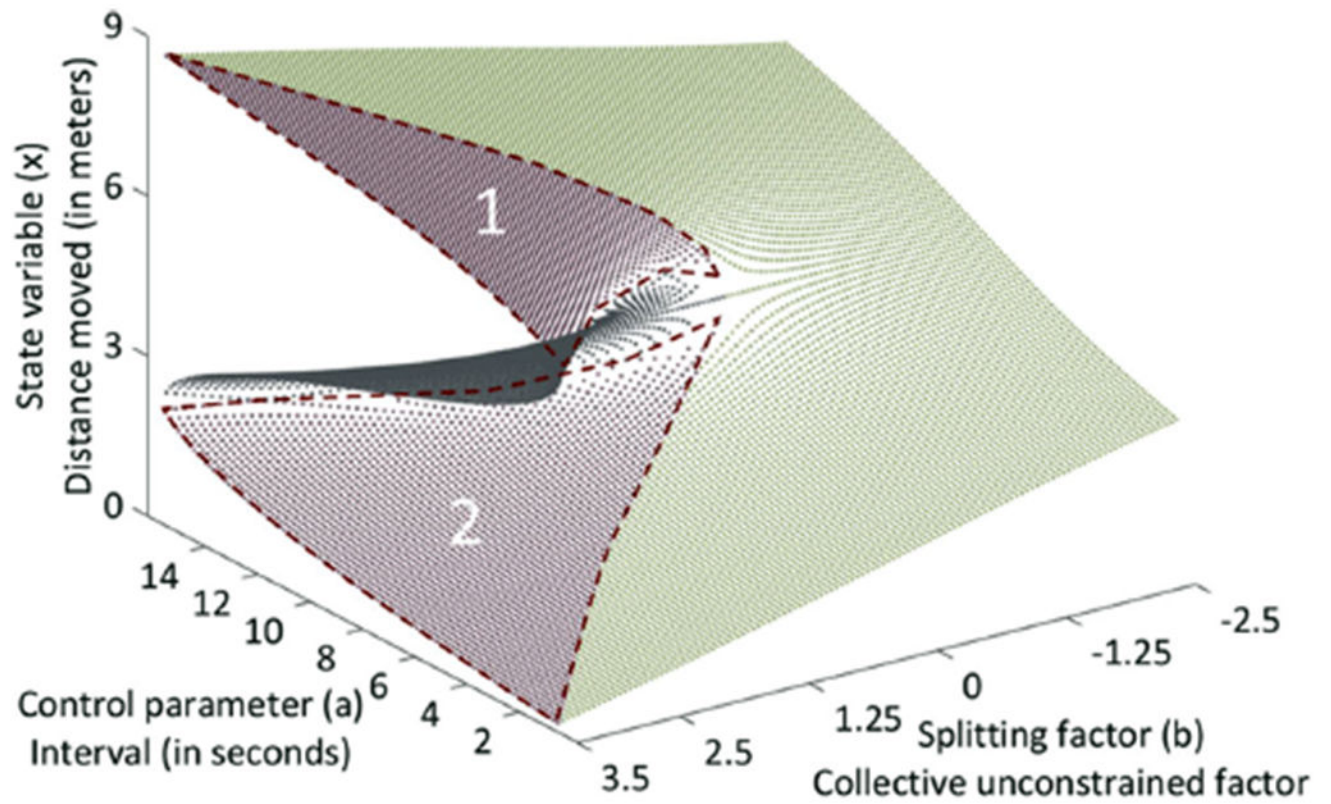


Figure 9.
Cusp Catastrophe solution surface with highlighted restricted subspaces for hard (marked '1') and soft (marked '2') constraint follow up experiments.

Table 1:

Distribution of type of dynamic per type of data.

Type of dynamic	Experimental	Simulated
stD_{small}	9.42%	8.57%
stD_{large}	35.51%	33.57%
$ptD_{gradual}$	23.91%	22.14%
$ptD_{nonlinear}$	31.16%	35.71%

Table 2:

Distribution of trajectories per follow-up experiment and type of data.

Type of dynamic	Experiment 2		Experiment 3	
	Experimental	Simulated	Experimental	Simulated
<i>stD_{small}</i>	25%	30%	6.67%	8.33%
<i>stD_{large}</i>	0%	0%	60%	53.33%
<i>ptD_{gradual}</i>	65%	60%	23.33%	28.33%
<i>ptD_{nonlinear}</i>	10%	10%	10%	10%